# E-Commerce Data Analysis – Project Brief Summary

This document summarizes the objective, scope, and key insights of an e-commerce data analysis project conducted using SQL and Power BI with AI. The analysis focuses on understanding business performance, customer behaviour, product demand, delivery efficiency, and payment trends using transactional order data.

## Project Topic

Analyse an e-commerce marketplace dataset to identify sales performance, customer purchasing patterns, product category trends, geographic demand, delivery performance, and payment method preferences.

## Analysis Objectives

- Understand overall business performance through revenue and order metrics.
- Identify top performing product categories.
- Analyse geographic distribution of customers and revenue.
- Evaluate delivery performance by comparing estimated and actual delivery dates.
- Study customer payment preferences across different payment methods.

## Dataset Overview

The dataset contains transactional e-commerce data including customers, orders, order items, products, and payments. It spans approximately two years of activity and includes over 100,000 orders. The dataset was imported into a PostgreSQL database where it was cleaned, structured, and prepared for analysis before being visualized in Power BI.

Key Findings

- **Total Revenue**: Approximately 16.1 million generated from delivered orders.
- **Total Orders**: Around 96,000 completed orders were analyzed.
- **Average Order Value**: Approximately 167 per order.
- **Top Product Categories**: Beauty & Health, Watches & Gifts, and Bed/Bath products generated the highest revenue.
- **Geographic Concentration:** Most orders originated from major Brazilian states such as São Paulo, Rio de Janeiro, and Minas Gerais.
- **Delivery Performance**: Around 92% of orders were delivered on time or earlier than estimated.
- **Payment Preference:** Credit cards accounted for the majority of transactions.

## Conclusion

This analysis demonstrates how raw transactional data can be transformed into actionable business insights through structured data cleaning, SQL analysis, and dashboard visualization. The project highlights the importance of data pipelines, data preparation, and analytical thinking in modern data analysis workflows